CS555, Spring 2026, HW 4
Due Friday, May 1, 5 PM.

In this homework, we consider the solution of the Reynolds bearing equation

$$-\nabla \cdot \left(\frac{h^3}{12\mu} \nabla p \right) \quad = \quad -\frac{1}{2} \nabla \cdot (\mathbf{u} h), \qquad p(x,y) = 0 \quad \text{on } \partial\Omega, \tag{1}$$

which is a variable-coefficient 2D Poisson problem governing the pressure distribution $p(x,y)$ between a bearing and a sliding surface moving at speed $\mathbf{u} = [u_x,\, u_y]$. Here, $h(x,y)$ is the prescribed bearing surface profile (height) and μ is the (constant) viscosity of the given fluid, which is air in this case. We will assume throughout that the velocity of the surface moving underneath the bearing is uniform, with value $\mathbf{u}(x,y) = [U,\, 0]$. In this assignment, we are modeling a read-write head supported above a spinning disk surface. A fuller description of the problem is provided in `hw_notes.pdf` on the Relate page.

For now, we consider the domain Ω to be the subset of the x-y plane denoted by the taper-flat *Slider 1* pad depicted in Fig. 1.[1]The upper panel of Fig. 1 shows the $L \times W$ domain comprising a leading *taper* region, $[0 : T] \times [-W/2 : W/2]$, followed by a *flat* region, $[T : L] \times [-W/2 : W/2]$. As seen in the lower panel, the profile of the taper region has a 1° relief with respect to the surface of the flat. The angle of the flat region is $\gamma := \tan^{-1}([h_1 - h_2]/L)$, where h_1 is the leading edge ($x = 0$) height of the bearing surface *in the absence of the taper* and h_2 is the trailing edge height at $x = L$. The unknown $p(x,y)$ corresponds to the pressure distribution between the bearing surface, which is stationary, and the spinning disk, which is moving with speed U. The prescribed gap between the slider and the disk is denoted as $h(x,y)$.[2]

You are provided with the FEM starter code, `taper_flat_a.m`, which solves (1) under the stated conditions. Parameters for a typical bearing configuration are defined in the code as follows

```
rho = 1.225;          % kg/m^3      density of air
nu  = 1.5e-5;         % m^2/s       kinematic viscosity
mu  = rho*nu;         %             dynamic viscosity
U   = 20;             % m/s         speed of plate
L   = .0041;          % m           slider length
W   = .15625*L;       % m           slider width (not used in 1D)
T   = .09375*L;       % m           slider taper length
Ta  = pi/180;         % m           Taper angle (1-deg)
h1  = 3.70e-7         % m           leading edge gap
h2  = 2.50e-7         % m           1/4-micron trailing edge gap
pitch = (h1-h2)/L;    %             leading edge height of slider
patm= 101325;         %             Pressure in N/m^2
n_per_gram= 9.81e-3; % N/g          Weight of a 1 gram mass
```

[1]Early slider designs were like a pair of skis, which allowed the Reynolds equation to be solved on two disconnected domains, *Slider 1* and *Slider 2*, whose relative position with respect to one another is fixed. Later designs, known as negative-pressure sliders, have significant bearing surfaces across the full slider.

[2]The read-write coil is mounted on the back end of the slider. Thus, from a design perspective, the value of h_2 under given loading conditions is the primary quantity of interest.

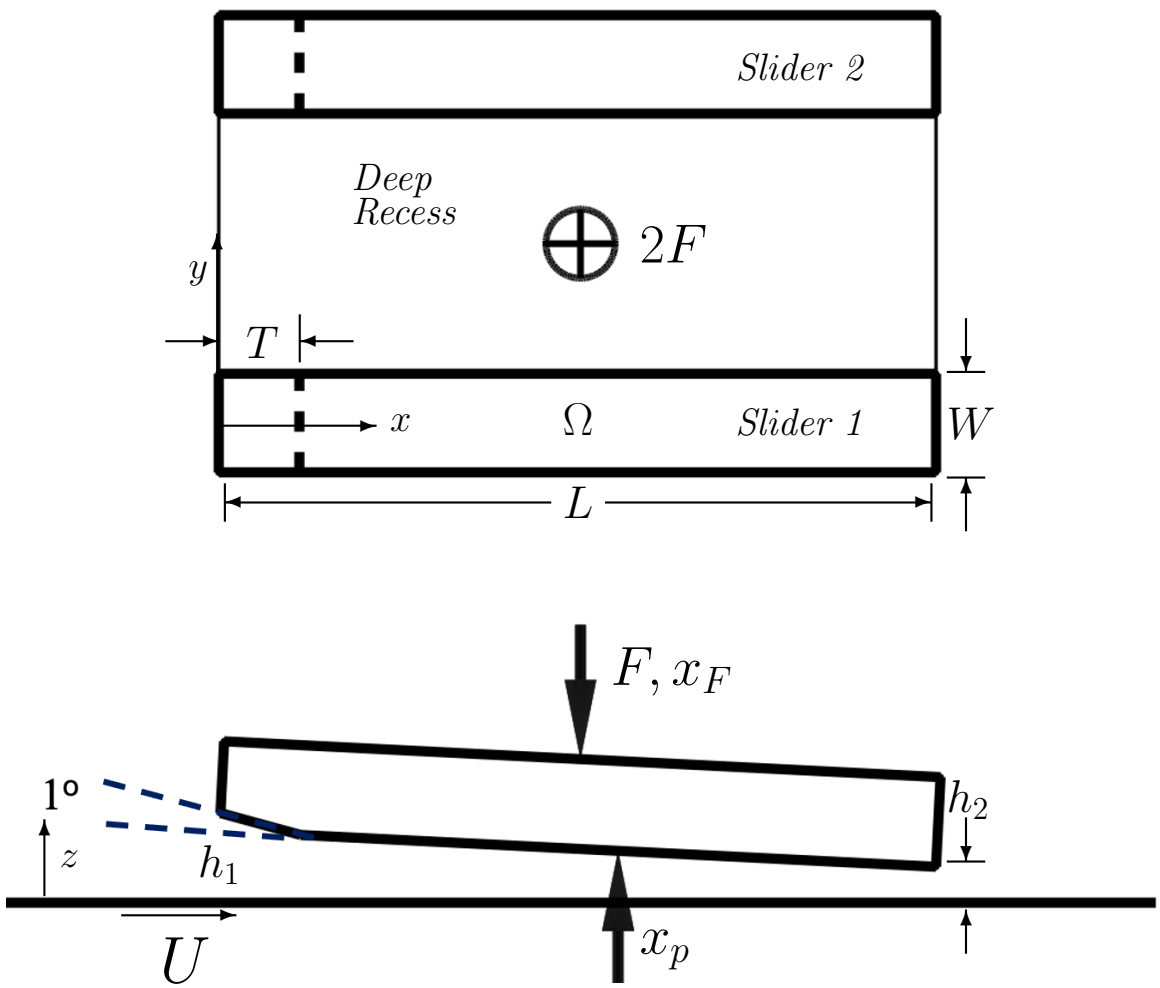

Figure 1: Taper-flat slider configuration: (top) plan-view for full slider pair with deep center recess, load $2F$ applied between the rails (sliders) at x-location x_F; (bottom) side-view for single slider with positive pitch, γ, load F, leading edge taper (1°), disk velocity U, and center-of-pressure x_p. The slider width is W, the length is L, and the taper length is T.

For **Part 1** of this assignment, answer the following.

1a. Add a few (< 10) lines of code to compute the load, F, and center-of-pressure, x_p:

$$F = \int_\Omega p(x,y)\,dx\,dy \qquad x_p = \frac{1}{F}\int_\Omega x\,p(x,y)\,dx\,dy. \tag{2}$$

(These integrals are easy in the FEM context.) To demonstrate that you are correctly computing F, verify your computed load against the analytical value $\tilde{F}$,

$$\tilde{F} \;=\; \frac{\Lambda\,L\,W}{(1-\alpha)^2}\left(\ln(\alpha) \,+\, 2\,\frac{1-\alpha}{1+\alpha}\right), \tag{3}$$

for the parameters and *wedge configuration* described in **1b**.

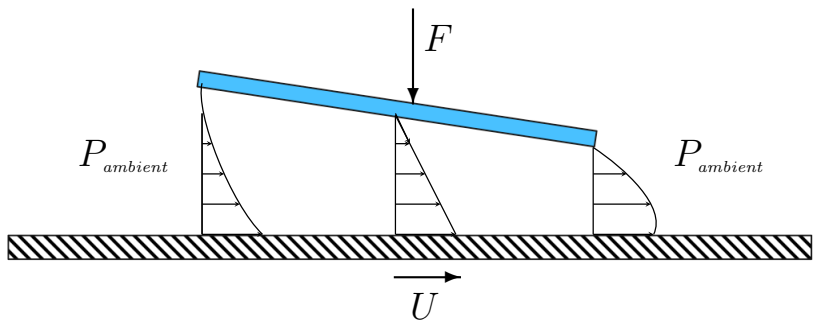

Figure 2: Flow configuration for a flat slider bearing.

1b. Verify the code against the *analytical solution* for the 1D wedge bearing problem (i.e., the "taper flat" bearing at a pitch γ, without the leading 1° taper, as illustrated in Fig. 2). Specifically, plot the maximum pointwise error in $p(x)$ vs. E_x on a log-log scale to demonstrate second-order convergence. (In this code, E_x is the number of elements in the x direction. Keep $E_y = 40$ for this case.) The analytical solution is

$$\tilde{p}(x) \quad = \quad \frac{\alpha\Lambda}{1-\alpha^2}\left(\frac{1}{H^2} - \frac{1}{\alpha^2}\right) \; - \; \frac{\Lambda}{1-\alpha}\left(\frac{1}{H} - \frac{1}{\alpha}\right), \tag{4}$$

where $\alpha := h_1/h_2$, $H(x) := \alpha + (1-\alpha)(x/L)$, and $\Lambda := 6\mu UL/h_2^2$. Note that h_1 is the leading-edge height in the absence of the taper. You will have to set your taper angle (`Ta`) to zero to conduct these tests. You will also need to convert the code to support Neumann conditions along the top and bottom edges in order to simulate a 1D (i.e., x-only) problem with the 2D code that you are verifying. This change can be made with a relatively straightforward modification to the setup of the restriction matrix, R.

Part 2: Compressibility. With peak pressures rising substantially above 1 atm, the air molecules are squeezed together, which implies that the fluid is denser between the disk and the slider than it is under ambient conditions. For the same reasons that the momentum equations are viscous dominated, the thermal transport is diffusion dominated and a reasonable assumption is that the temperature is uniform.[3] For air, we have the equation of state

$$p_a = \rho R T, \tag{5}$$

which is to say that $\rho = p_a \times$ *constant* for the isothermal model, where $p_a = p + p_{\text{atm}}$ is the *absolute pressure* and p_{atm} is the ambient pressure, taken to be 1 atmosphere = 101325 N/m^2 under normal operating conditions. (Different values are required for high altitudes.) Consequently, we modify the mass conservation equation (1) to include density, which is proportional to p_a.

We start with the mass conservation statement,

$$-\nabla \cdot \left(\frac{\rho h^3}{12\mu} \nabla p_a \right) = -\frac{1}{2} \nabla \cdot (\rho \mathbf{u} h), \qquad p_a(x,y) = p_{\text{atm}} \quad \text{on } \partial\Omega. \tag{6}$$

Note the **inhomogeneous** Dirichlet boundary condition for the absolute pressure, which must match the ambient (atmospheric) pressure, p_{atm} on $\partial\Omega$.

Using (5) with the isothermal model, we substitute p_a/RT for ρ and cancel $(RT)^{-1}$ from both sides to yield the *compressible Reynolds equation* for the *absolute pressure*, p_a,

$$-\nabla \cdot \left(\frac{p_a h^3}{12\mu} \nabla p_a \right) = -\frac{1}{2} \nabla \cdot (\mathbf{u} h p_a), \qquad p_a(x,y) = p_{\text{atm}} \quad \text{on } \partial\Omega, \tag{7}$$

which is nonlinear in p_a. More importantly, the unknown p_a also now appears on the right, which is the Couette-flow part of the mass conservation statement.

It is still convenient to split p_a into the pressure variance p (which we were computing in Part 1) and the constant atmospheric part, $p_{\text{atm}} = 101325 N/m^2$,

$$p_a(x,y) = p(x,y) + p_{\text{atm}}. \tag{8}$$

As before, we have

$$\frac{\partial p_a}{\partial x} = \frac{\partial p}{\partial x} \text{ and } \frac{\partial p_a}{\partial y} = \frac{\partial p}{\partial y}, \tag{9}$$

so $\nabla p = \nabla p_a$. For the Couette term on the right-hand side, however, we have

$$\frac{\partial}{\partial x} \left(\frac{Uh}{2} p_a \right) = \frac{\partial}{\partial x} \left(\frac{Uh}{2} p \right) + \frac{\partial}{\partial x} \left(\frac{Uh}{2} \right) p_{\text{atm}}. \tag{10}$$

The second term on the right leads to the same forcing term as in (1), albeit scaled by p_{atm}. The first term on the right is new. It is a linear (advection-like) term in the unknown p that corresponds to the local density (i.e., pressure) variation being propagated by the motion of the disk underneath the slider. (For our system, $\mathbf{u} = (U, V)$, but $V = 0$, so there is no corresponding $\frac{\partial}{\partial y}$ component.)

Rearranging (7) so that unknowns are on the left and known quantities are on the right yields the *compressible Reynolds equation*,

$$-\nabla \cdot \left(\frac{p_a h^3}{12\mu} \nabla p \right) + \frac{1}{2} \nabla \cdot (\mathbf{u} h p) = -\frac{p_{\text{atm}}}{2} \nabla \cdot (\mathbf{u} h), \tag{11}$$

$$p_a = p + p_{\text{atm}} \tag{12}$$

$$p = 0 \quad \text{on } \partial\Omega. \tag{13}$$

[3] The isothermal assumption does *not* hold for acoustic waves, where the state is *isentropic* rather than isothermal. There, we have $p = C\rho^\gamma$, with γ being the specific heat ratio.

The weighted residual formulation for this problem is *Find* $p \in X_0^N$ *such that for all* $v \in X_0^N$,

$$\int_\Omega \nu \nabla v \cdot \nabla p \, dV - \frac{1}{2} \int_\Omega \nabla v \cdot (\mathbf{u} h p) \, dV \quad = \quad \frac{1}{2} \int_\Omega \nabla v \cdot (\mathbf{u} h p_{\text{atm}}) \, dV, \tag{14}$$

with

$$\nu \quad := \quad (p_a) h^3/(12\mu) = (p + p_{\text{atm}}) h^3/(12\mu). \tag{15}$$

The matrix formulation for (14) is

$$R(\bar{A} - \bar{C}) R^T \underline{p} \quad = \quad R \bar{C} \underline{\bar{p}}_{\text{atm}}, \tag{16}$$

where $\bar{A} = \bar{A}(p_a)$ depends on p. Here, $\underline{\bar{p}}_{\text{atm}}$ is simply p_{atm} times the vector of all ones. An easy (and, in this case, convergent) strategy for treating the nonlinearity in $\bar{A}$ is to simply iterate (16), setting $p_a = p + p_{\text{atm}}$ and $\bar{A} = \bar{A}(p_a)$ after each iteration. With $p_a = p_{\text{atm}}$ as an initial guess, one needs only 5-6 iterations of (16) for the nonlinear solution to converge.

2a. Answer the following for the taper-flat configuration of (**1a**) using the compressible system (14)–(16):

- What is the trailing height h_2 (m)? (This is given to you.)
- What is the pitch, γ (radians)? (This is given to you.)
- What is the load, F (N)?
- What is the center-of-pressure, x_p (m)?

2b. Plot, side-by-side, the pressure distributions for the three configurations (**1a**), (**1b**), and (**2a**). Adjust the scales so that the distributions are clear and comment on the range of the vertical scale for each. The plots should be similar to Fig. 2 (left) in `hw_notes.pdf` and can be made with `trimesh` routine as done in the given starter code, `taper_flat_a.m`.

2c. Provide a table with F and x_p/L for each of the three cases and comment on your findings. (*Note:* in the compressible case, F should be computed using the *relative* pressure, p, not the absolute pressure, p_a. Why?)